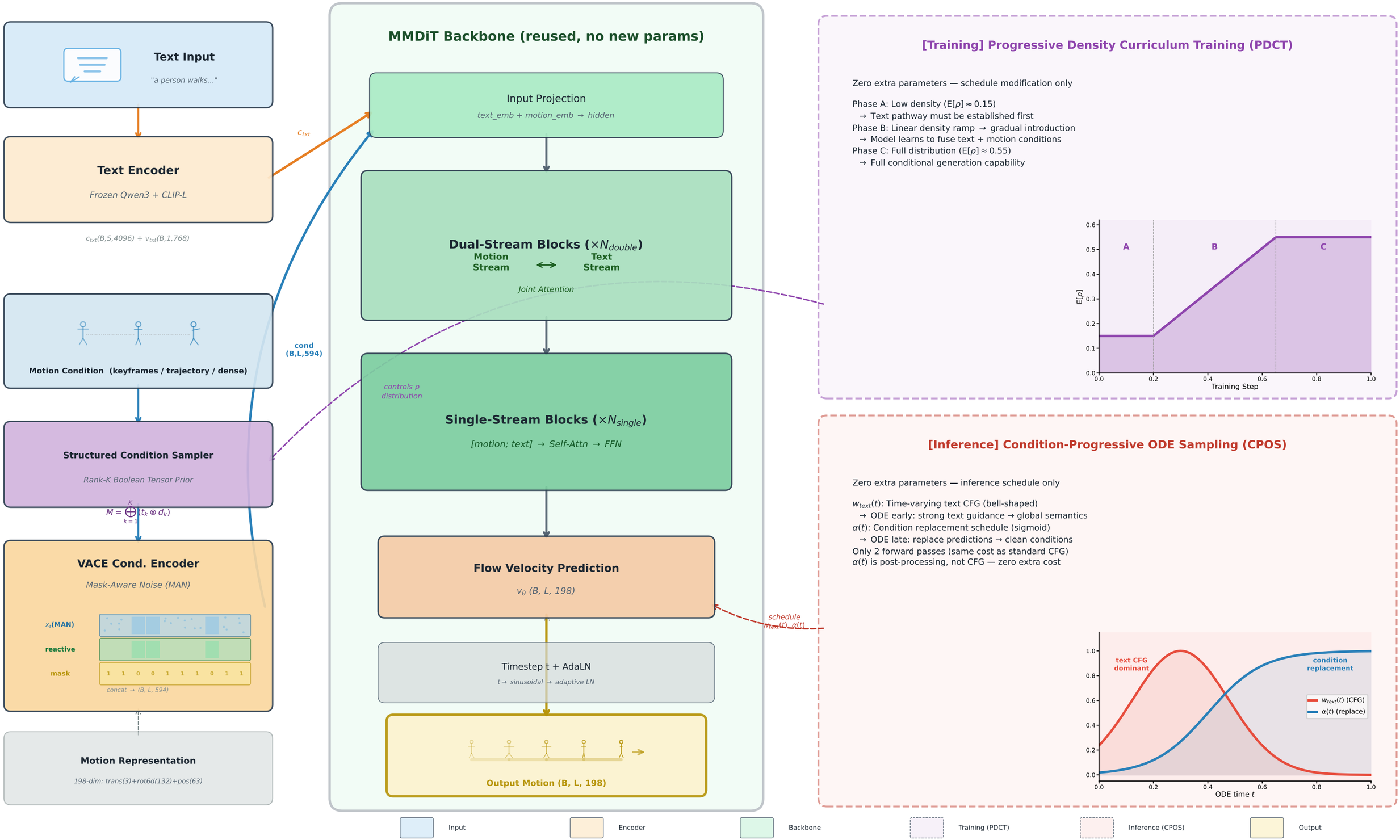

Text Input
"a person walks..."
Text Encoder
Frozen Qwen3 + CLIP-L
$c_{txt}(B,S,4096)$ + $v_{txt}(B,1,768)$
c_{txt}
Motion Condition (keyframes / trajectory / dense)
cond (B,L,594)
Structured Condition Sampler
Rank-K Boolean Tensor Prior
$M = \bigoplus_{k=1}^{K} (t_k \otimes d_k)$
VAE Cond. Encoder
Mask-Aware Noise (MAN)
x_t(MAN)
reactive
mask
1 1 0 0 1 1 1 0 1 1
concat → (B, L, 594)
Motion Representation
198-dim: trans(3)+rot6d(132)+pos(63)
MMDiT Backbone (reused, no new params)
Input Projection
text_emb + motion_emb → hidden
Dual-Stream Blocks (×N_{double})
Motion Stream
Text Stream
Joint Attention
controls ρ distribution
Single-Stream Blocks (×N_{single})
[motion; text] → Self-Attn → FFN
Flow Velocity Prediction
v_θ (B, L, 198)
Timestep t + AdaLN
t → sinusoidal → adaptive LN
Output Motion (B, L, 198)
schedule $w_{text}(t)$, $\alpha(t)$
[Training] Progressive Density Curriculum Training (PDCT)
Zero extra parameters — schedule modification only
Phase A: Low density (E[ρ] ≈ 0.15)
→ Text pathway must be established first
Phase B: Linear density ramp → gradual introduction
→ Model learns to fuse text + motion conditions
Phase C: Full distribution (E[ρ] ≈ 0.55)
→ Full conditional generation capability
A
B
C
E[ρ]
Training Step
[Inference] Condition-Progressive ODE Sampling (CPOS)
Zero extra parameters — inference schedule only
$w_{text}(t)$: Time-varying text CFG (bell-shaped)
→ ODE early: strong text guidance → global semantics
$\alpha(t)$: Condition replacement schedule (sigmoid)
→ ODE late: replace predictions → clean conditions
Only 2 forward passes (same cost as standard CFG)
$\alpha(t)$ is post-processing, not CFG — zero extra cost
text CFG dominant
condition replacement
$w_{text}(t)$ (CFG)
$\alpha(t)$ (replace)
ODE time t
Input
Encoder
Backbone
Training (PDCT)
Inference (CPOS)
Output